Is the remaining contingency sufficient?

Sankha Bhattacharjee | Hypothetical £5m office refurbishment and fit-out

Valuation date: 31 October 2026. Prepared: 5 November 2026. GBP, excluding VAT.

Management recommendation

Confirm the pending MEP design scope and priced instructions before committing further reserve. Escalate the quantified exposure to the sponsor. If the sponsor chooses a P80 funding basis, the model indicates a £53,631 gap that requires evidenced scope reduction, risk mitigation or additional funding.

Forecast measure	Amount
Approved funding	£5,000,000
Known forecast before residual risk	£4,915,000
Residual risk allowance, expected monetary value	£82,400
Expected forecast final cost	£4,997,400
Expected headroom	£2,600
P80 forecast / funding gap	£5,053,631 / £53,631
Modelled probability of exceeding funding	45.0%

Why £250,000 of unallocated contingency is not all available

Approved changes transfer £250,000 from the original £500,000 reserve into package budgets. The current allocated budget is £4.750m. The known forecast of £4.915m is £165,000 above that allocation, leaving only £85,000 within the funding cap for residual risk. Its expected allowance is £82,400, leaving £2,600 headroom.

Commercial priorities

Owner	Action and decision implication
MEP lead and QS	Validate the £180,000 ventilation revision. MEP carries £155,000 of the £165,000 net package forecast overrun.
Sponsor	Confirm the required funding confidence. P80 costs £5.054m under the stated risk model; expected cost alone gives limited protection.
Design and procurement	Validate the £30,000 potential door-finish credit. Even if approved at that value with risks unchanged, approximately £23,631 of P80 exposure remains.
QS and package managers	Reconcile £245,000 of accruals to completed work and certificates. Certification reclassification should not create extra final cost.

Status: synthetic worked example. No actual client costs, completed mitigation or achieved savings are represented.

Method and controls

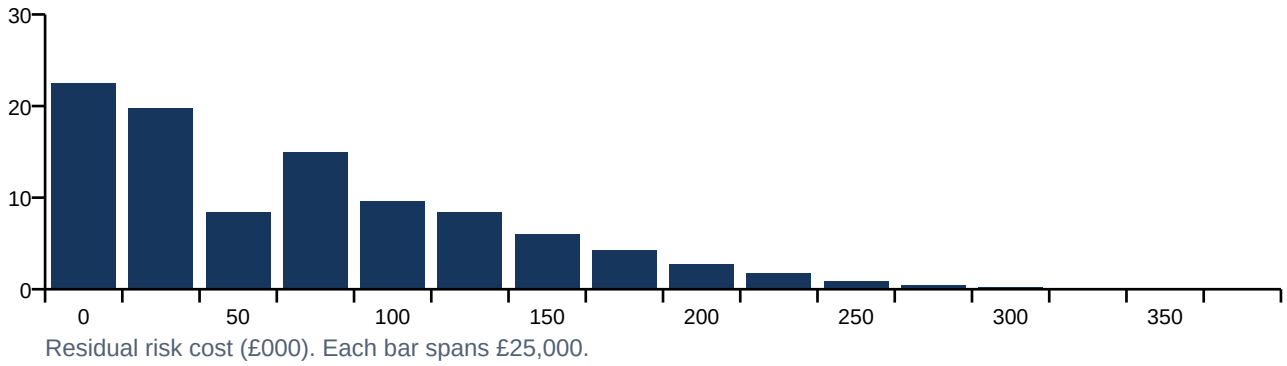
A forecast that reconciles commitments and cost recognition

Known forecast component	Amount
Certified gross work valuations	£1,930,000
Uncertified accruals	£245,000
Outstanding current contract commitments	£2,205,000
Unlet quantity × rate scope	£165,000
Known base-scope gaps	£55,000
Pending known changes, full best estimates	£315,000
Known forecast before residual risk	£4,915,000

Current contract value is £4.380m. Actual accrued cost of £2.175m plus remaining commitments of £2.205m reconciles to that value. Certified amounts are before retention and are not cash payments. Pending changes move into commitments once approved; the same scope is not added twice.

Risk model and reporting discipline

Three active event risks use independent Bernoulli occurrence and triangular cost impacts. Future delay is one mutually exclusive distribution: 0 weeks at 55%, 4 weeks at 35%, or 8 weeks at 10%, valued at £12,000 per week. R04 is excluded because CO07 already carries its scope. Later instruction CO10 falls outside the valuation cutoff.



Simulation: 50,000 draws, seed 53526. P50 £4,988,484; P80 £5,053,631; P90 £5,091,953. P80 means 80% of simulated outcomes are no greater than that amount under these assumptions. Analytical expected cost is not labelled P50.

Checks, refresh and limitations

Seventeen Python tests cover reconciliations, change approval, accrual reclassification, cutoff dates, opportunity exclusions, risk overlap and reproducibility. Excel also checks native sensitivity results and hides stale Monte Carlo outputs when tracked drivers change. Live Excel edits update the deterministic forecast; Python produces refreshed simulation results from the JSON inputs.

The case excludes land, professional fees, VAT and financing. All amounts, probabilities and impacts are assumptions. Independent risks and fixed known costs may understate correlated or unlisted exposures. This model demonstrates cost-reporting judgement and does not claim full professional-standard compliance.

Method references: IPA Cost Estimating Guidance (2021), on estimate ranges and risk allowances; RICS Cost reporting, first edition (2015), a historical reference for incurred, forecast and risk cost reporting.